

Statistics

Covering

- Classifying statistical data
- Interpreting statistical data
- Averages and measures of spread
- Statistical charts and diagrams
- Cumulative frequency diagrams O levels and Extended
- Histograms O levels and Extended

Data types

There are 2 methods data will be represented.

Ungrouped data is when data is not grouped and no frequency table is given.

Grouped data is when data is grouped with frequency table. This further breaks into 2. Discrete data, where data is given as integers, and Continuous data, where data is given as intervals, e.g 5 – 10.

Mean

For Ungrouped data its $a = \frac{\sum x}{n}$, Mean = $\frac{\text{Sum of values}}{\text{No. of values}}$.
 $an = \sum x$, Mean $\times$ No. of values = Sum of values.

For Discrete data its $a = \frac{\sum fx}{\sum f}$, Mean = $\frac{\text{Sum of multiplying corresponding } f \text{ and } x \text{ values}}{\text{Sum of frequency}}$

For Continuous data, we take x as the middle value of the intervals, then using the same
 $a = \frac{\sum fx}{\sum f}$.



Median

Median = Middle

For Ungrouped data, we first put them in order from small to large

If we have an odd number of values, we take the middle value.

If we have an even number of values, we take the average/mean of the middle 2 values.

For Discrete data its 1. Find $\frac{\sum f}{2}$, round it up if its decimal, 2. Add up frequencies till we reach a number $\geq$ our $\frac{\sum f}{2}$, the corresponding x is the median then.
E.g for the table, find the median value.

x	0	1	2	3	4
f	9	6	7	16	12

$\frac{\sum f}{2} = 25$. $0 \xrightarrow{+9} 9 \xrightarrow{+6} 15 \xrightarrow{+7} 22 \xrightarrow{+16} 38$. We see that 38 is above 25, hence we find that median is 3.

For Continuous data Median is found by midpoint of frequency of plotted data in a graph. e.g on a cumulative frequency graph

Mode

Mode = Modal = Common

For Ungrouped, its the value which repeats the most. If 2 or more are repeated the same number of times, they are also mode.

For Discrete its the value of x with the highest frequency.

Mode in Continuous data is called "modal class", the class with the highest frequency.

Range

Largest value of data – Smallest value of data



Unknown values in frequency table

We can find unknown frequency, x for cases.

x	0	1	2	3	4	5	6
f	6	12	x	7	4	10	8

If mode is given then to find the largest value of x its $x = \text{Mode} - 1$. For the range/possible values of x its $x \leq \text{Mode} - 1$. E.g For the given table $x = 11$, $x \leq 11$

If mode is x then to find the smallest value of x its $x = \text{Largest frequency given} + 1$. For the range/possible values of x its $x \geq \text{Largest frequency given} + 1$. E.g for the given table $x = 13$, $x \geq 13$

If median is given then to find greatest value of x its to 1. put your finger on the median column,

2. Use the formula Sum of Previous/Left frequencies = Sum of Next/Right frequencies.

E.g we have the median be 3, then:

1. Put your finger on median column

x	0	1	2		4	5	6
f	6	12	x		4	10	8

2. Add left and right frequencies and equate them

$$6 + 12 + x = 4 + 10 + 8$$

$$x = 4$$



Representation of data

There are 4 ways to represent data:

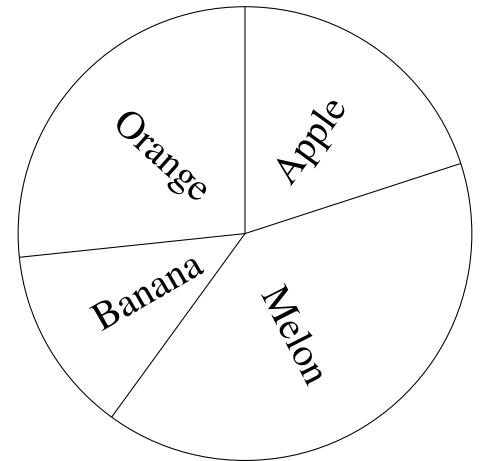
Pie Chart

Representation of data as sectors of a circle

Angle is found by $A = \frac{\text{Part}}{\text{Total}} \times 360$

E.g

Apple	3	$\frac{3}{15} \times 360 = 72$
Orange	6	$\frac{6}{15} \times 360 = 144$
Banana	2	$\frac{2}{15} \times 360 = 48$
Melon	4	$\frac{4}{15} \times 360 = 96$



Bar Graph

To draw, all bars must be of the same width. The gap between each bar must be the same. The height is the frequency for that corresponding x value. The x -axis has the first column, x values and the y -axis has the frequency column.

Frequency Table

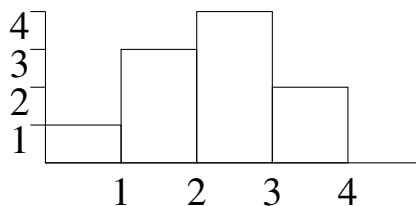
Sort/Count how many of x are in the Ungrouped data. For the Tally, it is a base 5 system where for values 1-4, you have group of 1-4 vertical lines. for 5 its |||| . After this you start again, 1-4 lines and a line in between for the 5. This is the Tally system.

Histograms

Bar graphs but no gaps. These are for Continuous data.

There are 2 cases

For equal Class Width, interval length, take the first column, x on the x -axis and take frequency on the y -axis. Make bars of equal width with no gaps and height of corresponding frequency for that interval.



Histograms

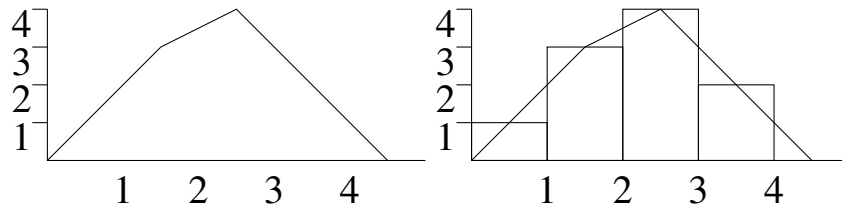
For variable Class Width, we have the width of bars be from Lower Boundary to Upper Boundary of the class, and the height is Frequency Density.

$$\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}.$$

$$\text{Class Width} = \text{Upper Class Boundary} - \text{Lower Class Boundary}$$

Frequency Polygon

Take the mid value of class intervals, x and then plot them (x, f) . x -axis is the mid points x and y -axis is the frequency. Join the points then. Histogram, midpoint of top edge.



Stem and leaf diagram

Core and Extended A stem and leaf graph is a table where each data value is split into a 'stem' (the first digit(s)) and a 'leaf' (usually the last digit). The key will be given.

E.g for the data

15 16 21 23 23 26 26 30 32 41

Stem	Leaf
1	5, 6
2	1, 3, 3, 6, 6
3	0, 2
4	1

Key: 2|3 represents 23



Cumulative Frequency

O levels and Extended

This is where you sum up frequencies as we go on. For a table of frequency, the cumulative frequency is filled by having the first frequency is same, next frequency by is add the previous frequency to it.

E.g

x	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
f	9	6	7	16	12

Cumulative

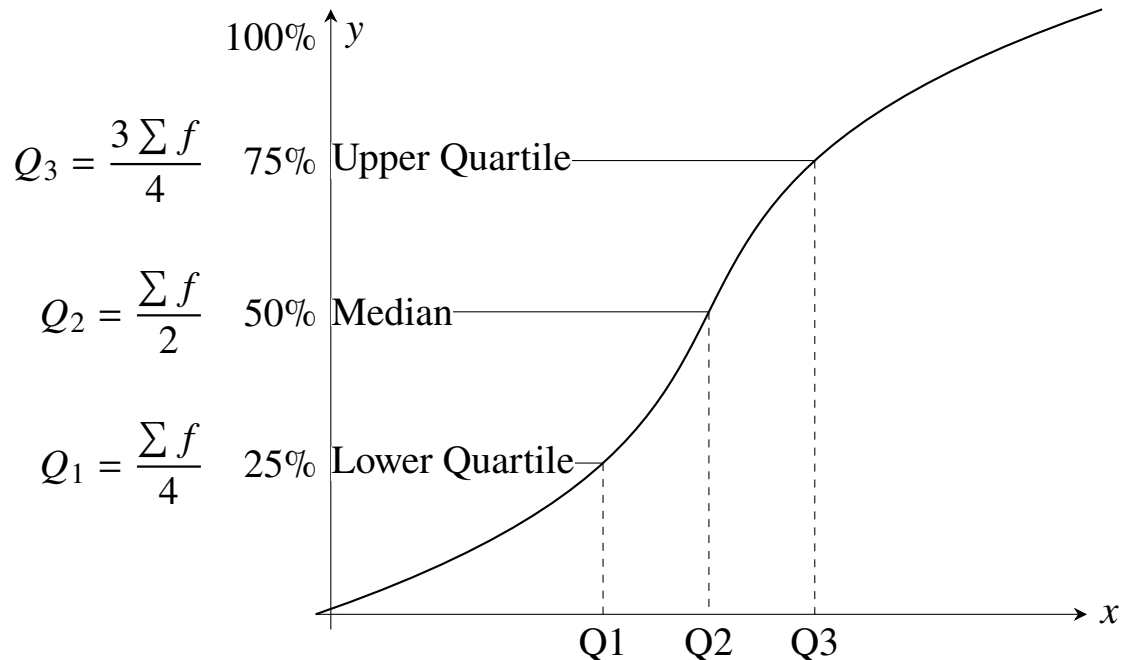
x	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
f	9	15	22	38	50

$\xrightarrow{+6}$ $\xrightarrow{+7}$ $\xrightarrow{+16}$ $\xrightarrow{+22}$

Quartiles

O levels and Extended

In a cumulative graph, there are 4 parts. y axis is Frequency.



$$x\text{th Percentile} = x\% \text{ of total frequency} = \frac{x}{100} \times \sum f.$$

$$\text{Interquartile} = Q_3 - Q_1$$

Patreon

Haris Rehman

Youtube

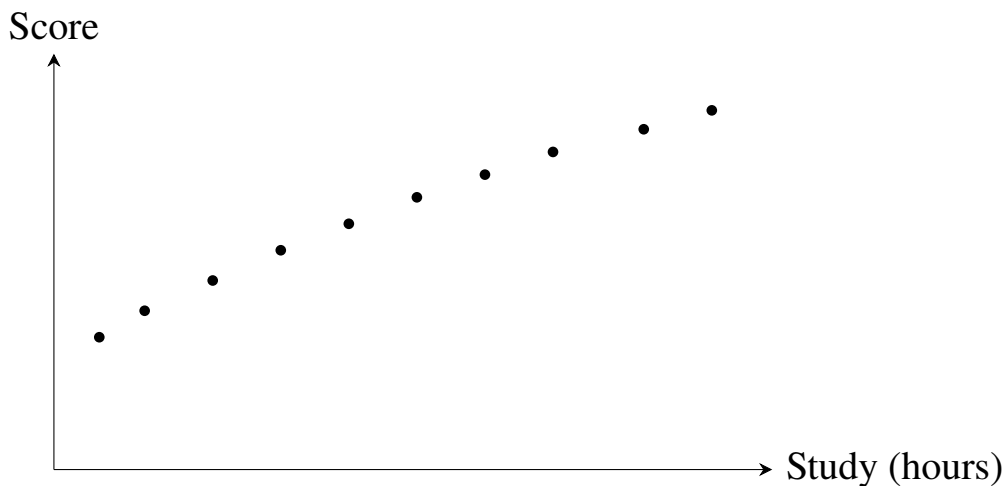


Scatter Diagrams

Covering

- Scatter diagrams

Scatter Diagrams A scatter diagram is plotting the values of 2 data lists. This is used to figure out cause and effects in the real world. Scatter diagrams are used to help get the Correlation between a cause and effect, e.g Hours spent studying is cause and effect is Score on result card. We can plot Hours spent studying on x and Score on result card on y . Using it we can see its a generally direct relation.



Correlation

Correlation is a description of the relationship between variables.

Correlation is said to be *positive* if both variables increase, e.g a tall girl usually takes a larger shoe size than a short girl. As one variable, height, increases the other variable, shoe size, also increases.

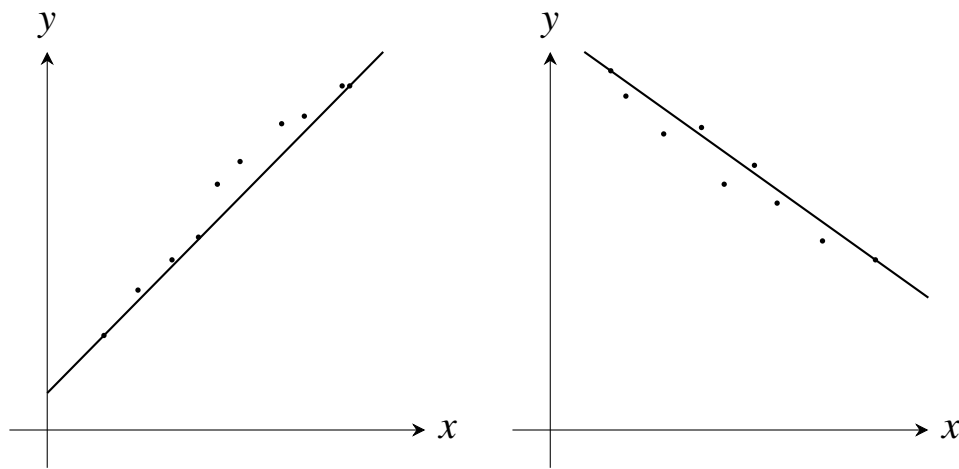
Correlation is said to be *negative* if one variable increases and the other variable decreases. e.g older men generally have less hair on their heads than younger men. As one variable, age, increases the other variable, amount of hair on head, decreases.



Best Line of Fit

A line of best fit is a straight line drawn through the centre of a group of data points on a scatter plot. The line of best fit shows whether these two variables appear to be correlated and can be used to help identify trends occurring within the dataset.

To draw it, use a ruler and make sure it goes through any 2 points, and have equal numbers of points above and below the line. Try to use points with easy coordinates to round/use.



Here in the examples I connected the first and last points, this is a good Best Line of Fit.

To find the equation of the drawn line, find the gradient (Coordinate Geometry) and extend your line to cross the y-axis for the y-axis.

Questions which ask about estimation using the Best Line of Fit, we first make a line from the value asked to the drawn line, then drop to the second axis.

